

Technology Stack

Date	01 October 2026
Team ID	SWTID-2026-9629
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

The technologies chosen for each layer of PocketSmart AI, with the reason for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (HTML/CSS, Jinja2)	HTML5, CSS3, JavaScript, Jinja2 templates	Lightweight, responsive forms and card layouts for the Home, Party and Jewelry planners; Jinja2 renders AI results server-side and integrates directly with FastAPI.
2	Backend / Server-Side (Python)	Python, FastAPI, Uvicorn	Fast, async-friendly framework with clean routing (/generate-home, /generate-party, /generate-jewelry), built-in request validation, CORS support and easy JWT authentication.
3	Database / Data Storage	SQLite	Stores user accounts, session data and recommendation history that power /history and the dashboard.
4	Cloud / Hosting / Deployment	Flask Development Server (Local Ubuntu/WSL)	Simple local run for development and demo; can be containerised later for cloud hosting.
5	Version Control & CI/CD	Git and GitHub	Team collaboration, change history and a shareable repository link for submission.
6	Third-Party APIs / Other Tools	Google Gemini 1.5 Flash Pro API; python-dotenv; Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO (mock / simulated data)	Gemini provides multimodal budget understanding and recommendation generation (text + outfit image); python-dotenv keeps API keys out of code; platform data supplies the product and service suggestions.